Supplementary Material: MAB-LDA

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Appendix

A. Modified ELBO

Standard variational LDA takes expectations under q. MAB-LDA instead takes expectations under the Gibbs posterior p and uses $\boldsymbol{\rho}$ as the variational parameter for the word-topic factor:

$$\mathcal{L}(\boldsymbol{\rho}) = \mathbb{E}_p[\log p(z|\theta)] + \mathbb{E}_p[\log p(\theta|\boldsymbol{\alpha})] + \mathbb{E}_p[\log p(\phi|\boldsymbol{\beta})] + \mathbb{E}_p[\log p(w|z,\phi)] - \mathbb{E}_p[\log q(z|\boldsymbol{\rho})] \tag{1}$$

B. Derivation of the Fixed Point (Proposition 1)

Expanding $\mathbb{E}_p[\log q(z|\boldsymbol{\rho})]$ using the Dirichlet log-normaliser and the digamma identity $\mathbb{E}[\log z_{vk}] = \psi(\rho_{vk}) - \psi(\sum_{k'} \rho_{vk'})$, then replacing the expectation under p with the Gibbs posterior estimate $\phi^T_{v,k}$:

$$\mathbb{E}_p[\log q] = \sum_v \Big[\log\Gamma(\textstyle\sum_k \rho_{vk}) - \sum_k \log\Gamma(\rho_{vk}) + \sum_k(\rho_{vk}-1)\big(\psi(\phi^T_{vk}) - \psi(\sum_{k'}\phi^T_{vk'})\big)\Big]$$

Differentiating $\mathcal{L}$ with respect to $\rho_{v,k}$:

$$\frac{\partial \mathcal{L}}{\partial \rho_{v,k}} = -\psi(\textstyle\sum_{k'}\rho_{vk'}) + \psi(\rho_{vk}) - \psi(\phi^T_{vk}) + \psi\big(\sum_{k'}\phi^T_{vk'}\big) = 0$$

By strict monotonicity of $x \mapsto \psi(x) - \psi(\sum x)$, $\rho^*_{vk} = \phi^T_{vk}$, i.e., $\boldsymbol{\rho}^* = \boldsymbol{\Phi}^\top$. ■

C. Derivation of the ELBO Lower Bound (Proposition 2)

By Jensen's inequality applied to the concave log over the product of per-document likelihoods:

$$\log p(\mathbf{w}|\boldsymbol{\alpha},\boldsymbol{\beta}) \geq \sum_{n,i} \log p(z_{n,i}, \mathbf{z}_{-(n,i)}, \mathbf{w}; \boldsymbol{\alpha},\boldsymbol{\beta}) - N|d| = \mathcal{L}^\circ$$

Normalising by $N \cdot |d|$ and subtracting 1 preserves the inequality, confirming $\mathcal{L}(t)$ acts as a valid theoretical diagnostic. ■

D. Complexity Analysis

TABLE I
Per-sweep time complexity of MAB-LDA components.

Operation	Cost	Notes
Full conditional (Eq. 1)	$O(K)$/token	Two K-vector products
Gamma trick (Eq. 2)	$O(K)$/token	K draws + argmax
Herfindahl (Eq. 4)	$O(K)$/token	Squared sum
$\boldsymbol{\rho}$ update (Eq. 7)	$O(1)$/token	Scalar EMA
TD checkpoint	$O(KV)$/sweep	Amortised $O(1)$/token
Total / sweep	$O(N \cdot L \cdot K)$	Linear in N

E. Hardware and Runtime

All experiments run on standard CPUs without GPU acceleration. Experiments were conducted on a commodity workstation (Intel Core i9-13900K, 24 cores, 64 GB DDR5 RAM). MAB-LDA completes 400 Gibbs sweeps on SemEval-2014 Restaurant (N=3,041, K=5) in an average of **48.2 seconds**, confirming deployment without cloud GPU infrastructure. Table I summarises per-operation costs. The Herfindahl reward adds $O(K)$ work per token — identical asymptotic cost to the base full conditional.